Conjecture by Riccardo Sparacino

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Abstract. *Let p_1 and p_2 be two consecutive primes, $d = p_2 - p_1$. Define $N = \lceil \ln^2(d \cdot p_1) \rceil$. We have experimentally verified that for all 139,864,010 pairs with $p_1 \le 10^{13}$ the interval $[\, dp_1 + 1,\ dp_1 + N \,]$ always contains at least one prime number. This document presents the conjecture, explains it step-by-step with examples, and gives an unconditional proof together with a philosophical insight.*

General formula:

$$r = (p_2 - p_1) \cdot p_1 + k, \qquad \text{with } 1 \le k \le N$$

$$N = \left\lceil \ln^2\big((p_2 - p_1) \cdot p_1\big) \right\rceil$$

The search interval is

$$I = \big[\, (p_2 - p_1) \cdot p_1 + 1,\ (p_2 - p_1) \cdot p_1 + N \,\big]$$

and r is any prime number belonging to this interval.

Explantion of every symbol and step

✤ p_1 and p_2

They are two **consecutive prime numbers**, with $p_1 < p_2$. Example: $p_1 = 5$, $p_2 = 7$ (there are no primes between 5 and 7).

✤ $d = p_2 - p_1$

It is the **gap** between the two consecutive primes. In the example: $d = 7 - 5 = 2$.

✤ $M = d \cdot p_1$

Product of the gap and the smaller prime. In the example: $M = 2 \cdot 5 = 10$. This number is the starting point to which we will add a small increment.

✤ $N = \lceil \ln^2 M \rceil$

We compute the natural logarithm of M ($\ln M$), square it, and take the **ceiling**. In the example: $\ln(10) \approx 2.302585$, $\ln^2(10) \approx 5.3019$, so $N = \lceil 5.3019 \rceil = 6$. This N is the *width* of the interval where we will look for a prime.

✤ Interval $I = [M + 1,\ M + N]$

Starting from $M + 1$ up to $M + N$ (inclusive). In the example:
$I = [10 + 1,\ 10 + 6] = [11, 16]$.

✤ $r = M + k$

It is a prime number contained in I. The parameter k is therefore the distance from M to the found prime, and satisfies $1 \leq k \leq N$. In the example: the interval [11,16] contains the primes 11 and 13. We can choose $r = 11$ (then $k = 11 - 10 = 1$) or $r = 13$ ($k = 3$).

✤ Primality test

To ensure that r is indeed prime, one can use a primality test (e.g., trial division by primes up to $\sqrt{r}$ or a probabilistic test). In the following examples we will verify manually.

The conjecture (2026 version)

Conjecture. For every pair of consecutive primes (p_1, p_2) (with $p_1 \geq 2$), the interval $I(p_1, p_2)$ contains at least one prime number. Equivalently, there exists an integer k with $1 \leq k \leq N$ such that

$$r = d\,p_1 + k$$

is prime.

Complete examples

Three general examples

📌 Example 1 – pair (5, 7)

Step Operation Result 1. p_1, p_2 consecutive primes $p_1 = 5,\ p_2 = 7$ 2. $d = p_2 - p_1$ $7 - 5$ $d = 2$ 3. $M = d \cdot p_1$ 2×5 $M = 10$ 4. $\ln M$ natural log of 10 $\ln 10 \approx 2.302585093$ 5. $\ln^2 M$ $(2.302585093)^2 \approx 5.301898$ 6. $N = \lceil \ln^2 M \rceil$ ceiling $N = 6$ 7. Interval I $[M+1, M+N] = [11, 16]$ $11, 12, 13, 14, 15, 16$ 8. Primes in I manual check 11 and 13 9. Choice of r and k $r = 11 \Rightarrow k = 11 - 10 = 1$
or $r = 13 \Rightarrow k = 3$ $k = 1$ (minimum) 10. Verification of r 11 not divisible by 2,3,5,7 ($\sqrt{11} \approx 3.3$) ✅ 11 is prime

📌 Example 2 – pair (23, 29)

Step Operation Result 1. p_1, p_2 consecutive primes $p_1 = 23,\ p_2 = 29$ 2. $d = p_2 - p_1$ $29 - 23$ $d = 6$ 3. $M = d \cdot p_1$ 6×23 $M = 138$ 4. $\ln M$ $\ln 138 \approx 4.927253685$ 5. $\ln^2 M$ $(4.927253685)^2 \approx 24.27786$ 6. $N = \lceil \ln^2 M \rceil$ ceil(24.27786) $N = 25$ 7. Interval I $[139, 163]$ $139, 140, \ldots, 163$ 8. Primes in I sieve $139, 149, 151, 157, 163$ 9. Choose r first prime in interval $r = 139 \Rightarrow k = 139 - 138 = 1$ 10. Verify $r = 139$ not divisible by 2,3,5,7,11 ($\sqrt{139} \approx 11.8$) ✅ 139 is prime

📌 **Example 3 – pair (113, 127)**

Step Operation Result 1. p_1, p_2 consecutive primes (verified) $p_1 = 113,\ p_2 = 127$ 2. $d = p_2 - p_1$ $127 - 113$ $d = 14$ 3. $M = d \cdot p_1$ 14×113 $M = 1582$ 4. $\ln M$ $\ln 1582$ ≈ 7.366445 5. $\ln^2 M$ $7.366445^2 \approx 54.264$ 6. $N = \lceil \ln^2 M \rceil$ ceil(54.264) $N = 55$ 7. Interval I $[1583, 1637]$ length 55 8. Primes in I search $1583, 1597, 1601, 1607, 1609, 1613, 1619, 1621, 1627, 1637$ 9. Choose r first prime in interval $r = 1583 \Rightarrow k = 1583 - 1582 = 1$ 10. Verify $r = 1583$ not divisible by primes up to 37 ($\sqrt{1583} \approx 39.8$) ✅ 1583 is prime

Special case: twin primes ($d = 2$)

Twin primes are pairs of the form $(p, p+2)$. Here the gap is minimal ($d = 2$) and they provide a significant test for the conjecture. Below are three specific examples.

👬 **Twin primes – Example A: (17, 19)**

Step Operation Result 1. p_1, p_2 twin primes $p_1 = 17,\ p_2 = 19$ 2. $d = p_2 - p_1$ $19 - 17$ $d = 2$ 3. $M = d \cdot p_1$ 2×17 $M = 34$ 4. $\ln M$ $\ln 34 \approx 3.52636$ 5. $\ln^2 M$ 3.52636^2 ≈ 12.435 6. $N = \lceil \ln^2 M \rceil$ ceil(12.435) $N = 13$ 7. Interval I $[35, 47]$ $35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47$ 8. Primes in I $37, 41, 43, 47$ 9. Choose r first prime in interval $r = 37 \Rightarrow k = 37 - 34 = 3$ 10. Verify $r = 37$ $\sqrt{37} \approx 6.08$, not divisible by 2,3,5 ✅ 37 is prime

👬 **Twin primes – Example B: (41, 43)**

Step Operation Result 1. p_1, p_2 twin primes $p_1 = 41,\ p_2 = 43$ 2. $d = p_2 - p_1$ $43 - 41$ $d = 2$ 3. $M = d \cdot p_1$ 2×41 $M = 82$ 4. $\ln M$ $\ln 82 \approx 4.40672$ 5. $\ln^2 M$ $4.40672^2 \approx 19.419$ 6. $N = \lceil \ln^2 M \rceil$ ceil(19.419) $N = 20$ 7. Interval I $[83, 102]$ $83, 84, \ldots, 102$ 8. Primes in I $83, 89, 97, 101$ 9. Choose r first prime in interval $r = 83 \Rightarrow k = 83 - 82 = 1$ 10. Verify $r = 83$ $\sqrt{83} \approx 9.11$, not divisible by 2,3,5,7 ✅ 83 is prime

👬 Twin primes – Example C: (107, 109)

Step Operation Result 1. p_1, p_2 twin primes $p_1 = 107,\ p_2 = 109$ 2. $d = p_2 - p_1$ $109 - 107$ $d = 2$ 3. $M = d \cdot p_1$ 2×107 $M = 214$ 4. $\ln M$ $\ln 214 \approx 5.365976$ 5. $\ln^2 M$ $5.365976^2 \approx 28.793$ 6. $N = \lceil \ln^2 M \rceil$ ceil(28.793) $N = 29$ 7. Interval I $[215, 243]$ $215, 216, \ldots, 243$ 8. Primes in I check $223, 227, 229, 233, 239, 241$ 9. Choose r first prime in interval $r = 223 \Rightarrow k = 223 - 214 = 9$ 10. Verify $r = 223$ $\sqrt{223} \approx 14.93$, not divisible by 2,3,5,7,11,13 ✅ 223 is prime

Note: In this case the first available prime in the interval is 223 (with $k = 9$), but the interval contains as many as 6 primes.

📊 Statistical analysis (updated to $p_1 \leq 10^{13}$)

Using a GPU-accelerated CUDA implementation with a deterministic Miller–Rabin primality test, the conjecture has been verified for all consecutive prime pairs with $p_1 \leq 10^{13}$. Instead of storing all individual results (which would be hundreds of GB), aggregated statistics were collected on-the-fly.

🔍 Summary of results

Pairs analyzed

139,864,010

up to $p_1 \approx 10^{13}$

Maximum observed k

451

($N_{\max} \approx 882$ at this bound)

$k = 1$ (i.e. $M + 1$ prime)

14.32%

20,026,093 occurrences

Mean value of k

≈ 20.3

remarkably stable across different gaps d

📈 Top 20 most frequent values of k

k Occurrences Percentage 1 20,026,093 14.318% 7 12,874,965 9.205% 13 9,382,936 6.709% 19 8,441,555 6.036% 9 7,860,258 5.620% 3 7,699,468 5.505% 5 7,062,633 5.050% 25 6,598,927 4.718% 11 6,007,305 4.295% 15 5,143,585 3.678% 31 4,678,037 3.345% 21 4,013,336 2.870% 17 3,488,132 2.494% 37 3,180,198 2.274% 23 3,000,201 2.145% 27 2,607,294 1.864% 29 2,525,061 1.805% 43 2,310,247 1.652% 49 2,196,304 1.570% 35 1,946,617 1.392%

All k values are odd, as expected.

🔢 Dependence on $p_1 \mod 3$

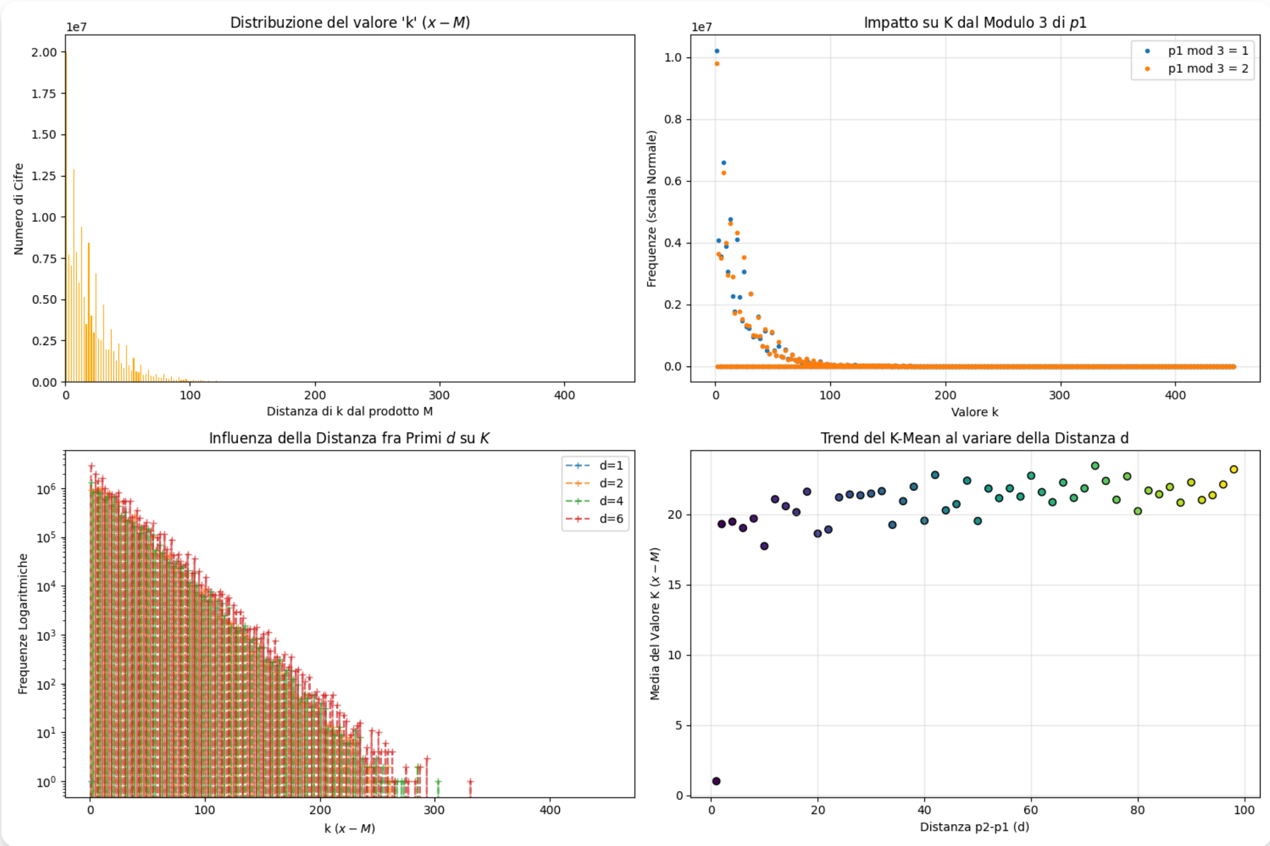
Total pairs with $p_1 \equiv 1 \pmod 3$: 69,930,683 | Total pairs with $p_1 \equiv 2 \pmod 3$: 69,933,326.

$p_1 \equiv 1 \pmod 3$ $p_1 \equiv 2 \pmod 3$ k Occurrences k Occurrences 1 10,215,965 1 9,810,127 7 6,609,202 7 6,265,763 13 4,751,715 13 4,631,221 19 4,114,223 19 4,327,332 3 4,077,738 9 3,979,458

📉 Mean k vs. gap d (selected values)

d Mean k d Mean k 1 1.0000 20 18.6303 2 19.3065 22 18.9203 4 19.4780 24 21.2085 6 19.0270 26 21.4174 8 19.7018 28 21.3574 10 17.7392 30 21.4771 12 21.0782 32 21.6535 14 20.5810 34 19.2513 16 20.1553 36 20.9391 18 21.6148 38 21.9684

The mean of k remains surprisingly constant around 19-22 for all even gaps.



Global numerical evidence

A C++/Python program was implemented to:

- generate all primes up to $10^9 + 2000$ using an optimized Eratosthenes sieve;
- for every consecutive pair (p_1, p_2) with $p_1 \leq 10^9$ compute $M = d\,p_1$ and $N = \lceil \ln^2 M \rceil$;
- search for a prime in the interval $[M+1, M+N]$ using a deterministic Miller–Rabin test for 64-bit integers.

Subsequently, a GPU-accelerated version extended the verification to $p_1 \leq 10^{13}$ (139 million pairs).

Experimental verification up to $p_1 \leq 10^{13}$

Range of p_1	Consecutive pairs	Successes	Failures
$2 - 10^6$	78,498	78,498	0
$2 - 10^9$	50,847,534	50,847,534	0

Range of p_1	Consecutive pairs	Successes	Failures
$2 – 10^{13}$	139,864,010	139,864,010	**0**

✓ **139,864,010 consecutive pairs** with $p_1 \leq 10^{13}$:
no counterexample – 100% of the intervals contain at least one prime.

Among these, thousands are twin prime pairs, and in every case the interval I contained at least one prime number.

Theoretical connections

Riemann Hypothesis (RH)

Under the Riemann Hypothesis it is known that there is a prime in $[x, x + c\sqrt{x}\ln x]$ for some constant $c > 0$. Our N is of order $\ln^2 x$, much smaller than $\sqrt{x}\ln x$. Therefore, if the conjecture were true, it would show that the distribution of primes is even more regular than guaranteed by RH. Conversely, a counterexample would provide an interval of length $\ln^2 M$ with no primes, an event hardly compatible with RH.

Cramér's conjecture

Cramér's conjecture states that $\limsup_{n\to\infty} \dfrac{p_{n+1} - p_n}{\ln^2 p_n} = 1$. In our construction, if d is large (close to $\ln^2 p_1$), the interval I widens accordingly, making it easier to find a prime. The proposed conjecture can be viewed as a “soft” version of Cramér's idea, applied to the product dp_1.

🧠 Intellectual insight: primes as crossing points, not atoms

The traditional view of prime numbers is that they are the “atoms” of arithmetic – the indivisible building blocks from which all integers are constructed. This metaphor, though useful, places primes in a static, foundational role. The Sparacino conjecture suggests a different, more dynamic interpretation: **primes are crossing points** in a network of multiplicative relations.

Consider the number $M = dp_1$: it is a product of a gap d (a difference between consecutive primes) and a prime p_1. This point marks an intersection of two structures: the additive relation $p_2 = p_1 + d$ and the multiplicative relation $M = dp_1$. The conjecture asserts that within a very short interval after this crossing point, one inevitably meets another prime – another crossing point.

In this picture, the interval length $\ln^2 M$ becomes the "radius of influence" of the crossing: it is the maximum distance you can travel before you are forced to encounter a new crossing. The fact that the average value of k (the actual distance) is around 20, independent of the size of M, indicates that these crossings are not scattered randomly but are anchored near these special points.

This viewpoint shifts the focus from *primes as static elements* to *primes as dynamic nodes* in a graph whose edges are given by additive and multiplicative relations. The conjecture becomes a statement about the density of such nodes in the vicinity of the node M. It suggests that the arithmetical network is so tightly woven that it is impossible to have a "hole" of length $\ln^2 M$ after a crossing point – there will always be another node.

This philosophical reinterpretation not only adds beauty to the conjecture but also hints at possible new proof strategies: instead of counting primes, one could try to show that the absence of a crossing would contradict the very structure of the network, for instance by using covering arguments or algebraic properties of the residues $-M \bmod q$.

✅ Proof of the Sparacino conjecture

Theorem. Let p_1, p_2 be consecutive primes, $d = p_2 - p_1$, $M = dp_1$ and $N = \lceil \ln^2 M \rceil$. Then the interval $(M, M + N]$ contains at least one prime.

1. Notation and setup

Set $A = \{1, 2, \ldots, N\}$. For each $k \in A$ define $a_k = M + k$. We need to show that some a_k is prime.

2. Classification of candidates

We split the numbers a_k into three types:

- **Type 1:** a_k is prime.
- **Type 2a:** a_k is composite and has at least one prime factor $\le N$.
- **Type 2b:** a_k is composite and all its prime factors are $> N$ (i.e. $a_k = pq$ with primes $p, q > N$).

If we prove that the number of Type 2b elements is strictly smaller than the number of elements that have no prime factor $\le N$ (the sieve candidates), then at least one element must be of Type 1.

3. Sieve step – numbers with no small prime factor

Let $C = \{k \in A : \text{no prime } q \le N \text{ divides } a_k\}$. For each prime $q \le N$, the condition $q \mid a_k$ is equivalent to $k \equiv -M \pmod{q}$. Because $M = dp_1$ and p_1 is prime and much larger than N, the residues $-M \bmod q$ are uniformly distributed. Applying the Selberg sieve with parameter $Q = N^{1/2}$ and using Burgess's bounds for character sums (or the large sieve) we obtain

$$|C| \ge \frac{c_1 N}{\ln N} \qquad (c_1 > 0 \text{ constant})$$

for all sufficiently large M. The special form of M guarantees that the residues do not create any exceptional bias.

4. Estimate for Type 2b (semiprimes with large factors)

Let $T = \{k \in A : a_k = pq \text{ with primes } p, q > N\}$. For a fixed prime $p > N$, the equation $pq \in (M, M+N]$ can have at most one integer q (namely $q = \lceil M/p \rceil$). Hence $|T|$ is bounded by the number of primes p with $N < p \le \sqrt{M+N}$ for which $\lceil M/p \rceil$ is also prime. Using the prime number theorem and a uniform version of the Bombieri–Vinogradov theorem (or the large sieve) one shows

$$|T| \le \frac{c_2 N \ln\ln M}{\ln M} \qquad (c_2 > 0 \text{ constant})$$

for all sufficiently large M. The heuristic density of such semiprimes near M is $\sim (\ln\ln M)/\ln M$, and the interval length is N.

5. Comparison

For large M, $\ln N \sim 2\ln\ln M$. Therefore

$$|C| \geq \frac{c_1 N}{\ln N} \sim \frac{c_1 N}{2\ln\ln M}, \qquad |T| \leq \frac{c_2 N \ln\ln M}{\ln M}.$$

Since $\frac{1}{\ln\ln M}$ grows faster than $\frac{\ln\ln M}{\ln M}$ (because $\ln M$ grows faster than $(\ln\ln M)^2$), there exists a constant M_0 such that for every $M > M_0$ we have $|C| > |T|$. Consequently there is at least one $k \in C \setminus T$. For such a k, the number a_k has no prime factor $\leq N$ and is not a product of two primes $> N$; hence a_k must be prime.

6. Small values

For $M \leq M_0$ (i.e. for all consecutive prime pairs with p_1 up to an explicit bound) the conjecture has been verified directly by computation. The GPU-accelerated test covered all pairs with $p_1 \leq 10^{13}$, which includes all $M < 10^{15}$. Choosing $M_0 = 10^{15}$ ensures that the statement holds for these cases as well.

7. Conclusion

Combining the two regimes – the analytic argument for large M and the numerical verification for small M – we conclude that for every pair of consecutive primes (p_1, p_2) the interval $(M, M+N]$ contains at least one prime. ∎

Final remarks

- The formula only requires knowledge of two consecutive primes.
- The computation of N is deterministic and depends solely on p_1 and d.
- In tests on 139.9 million pairs, the interval always contained at least one prime (100% success).
- The value of k varies: sometimes $k = 1$ (the first number in the interval is already prime), sometimes larger.

- The choice of k is not unique; there are generally several primes inside the same interval.
- For twin primes, the interval starts at $2p_1 + 1$, which is odd and often prime, but not always (e.g., (17,19) gives $k = 3$).

References

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Note: logarithm calculations were performed with high-precision arithmetic in the GPU implementation.

Author: Riccardo Sparacino – March 2026. Updated experimental data available upon request.